Week 2 Data Readiness Report: Customer Support Assistant

Item	Value
Student	Nguyen Tien Viet Hung
Course	QuanSkill AI Engineering and Deployment, Chapter 4 (pp. 287–402)
Track	Customer Support Assistant (same as Week 1)
Pipeline version	`week2-v1.0`, config SHA-256 `4c7e2228497ced92…` (full value in `data/processed/run_manifest.json`)
Data	**Synthetic.** Fictional retailer "ShopNova"; no real personal or company data
Decision	**READY WITH CONDITIONS** (section 12)

Evidence labels used in this report: **Observed** (measured in this dataset), **Estimated**, **Assumed**, **Unknown**. Every number marked *Observed* was computed by the pipeline or the notebook. Because the dataset is synthetic, an *Observed* value describes this dataset only. It is not evidence about a real support team.

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1. Project and intended AI task

Week 1 problem (unchanged). Tier-1 support agents re-answer repeated policy questions by searching scattered sources by hand, which is slow and inconsistent.

Intended AI task (Week 1 MVP). Given an agent's English-language question about return, refund, delivery, warranty, order-change, account or payment policy, retrieve the relevant **approved** knowledge passages and draft a short cited answer, or abstain, for the agent to review.

What Week 2 must deliver for that task

Dataset	Role in the next stage	Output
Knowledge	The **only** evidence source the assistant may cite	`data/processed/knowledge/knowledge_articles.jsonl`
Interactions	Realistic customer questions for **evaluation** (never used as evidence)	`data/processed/interactions/{train,validation,test}.jsonl`
Ticket outcomes	Business baseline B_0 (handle time) only. Post-outcome fields, never features.	`data/processed/interactions/ticket_outcomes.csv`

Out of scope for Week 2: modelling, embeddings, chunking, vector index, RAG, prompts, UI and deployment.

2. Raw-data inventory and provenance

Full inventory with SHA-256 hashes: `data/raw/SOURCE_INVENTORY.csv` (15 sources). All raw files are preserved unchanged. Gate **G1** re-hashes them on every run and stops the pipeline if any hash differs.

Source ID	Type	Location	Origin / owner	Collection	Obtained	Volume	Refresh	Role	Sensitivity	Known limitation
SRC-KB-001 … SRC-KB-014	Markdown + front matter	`data/raw/knowledge_base/KB-*.md`	ShopNova help-center CMS (synthetic); owner per article	CMS export, 1 file per article	2026-09-30	14 articles	on policy change; owner review yearly	knowledge evidence	internal, no personal data	approval and review dates are self-reported
SRC-TICKETS-2026-09-30	CSV	`data/raw/tickets/tickets_export_2026-09-30.csv`	ShopNova helpdesk (synthetic)	full export, tickets created 1 Jun – 29 Sep 2026	2026-09-30	706 rows, 12 columns	weekly (planned)	evaluation questions and reference replies	confidential if real (free text may hold personal and payment data)	mixed timestamp formats; labels optional; duplicated rows

How the data was produced. `src/data/generate_synthetic_data.py` (seed 2026) was run once. It injects realistic defects on purpose so that the readiness controls can be tested. The ticket CSV SHA-256 is `b9dae651…cb60b7`.

Stable identifiers. Sources: `SRC-…`. Articles: `kb_id` (KB-001…). Messages: `message_id` (MSG-000001…). Threads: `ticket_id` (TCK-10001…). Every record also carries `record_sha256`, the hash of its raw bytes or raw field values.

Lineage.

```
data/raw/tickets/tickets_export_2026-09-30.csv  (sha256 b9dae651…, row 223)
  └─ build_dataset.py week2-v1.0: parse → hash row → critical rules → whitespace/UTC/channel/label normalisation
       → PII redaction → thread rules → split (temporal, by ticket) → length fence (fitted on train) → outcome
```

```
        ├─ data/processed/manifest.csv            source_id=MSG-000223, record_sha256=79cbda68…, outcome=publish
        └─ data/processed/interactions/validation.jsonl  message_id=MSG-000223, source_row=223, source_sha256=79cbda68…
```

Trace example (Observed). The published record `MSG-000223` in `validation.jsonl` has `source_row = 223` and `source_sha256 = 79cbda68c34a…ff542`. The manifest row for MSG-000223 shows the same hash, file and row, with outcome `publish`. Re-hashing raw row 223 reproduces this value (README, "Trace one record"). The raw file hash matches the inventory (G1). The raw value `2026-08-05 15:56:00` (naive local time) became `2026-08-05T08:56:00Z`, and `delivery time` became `delivery_time`.

3. Raw profiling baseline (before any transformation)

Source: `notebooks/02_data_profiling.ipynb`, Part 1 (HTML export alongside the notebook).

Measure	Value (Observed)
Ticket export size	706 rows × 12 columns, all loaded as text
Distinct `message_id`	694 of 706, so 12 duplicate rows
Distinct tickets	230, plus 4 rows with blank `ticket_id`
Blank `message_text` (after whitespace trim)	6 (0.8%)
`created_at` formats	ISO-UTC 428 (60.6%), +07:00 offset 177 (25.1%), **naive** 96 (13.6%), **invalid 5 (0.7%)**; **2 future**
`channel` spellings	11 spellings for 3 channels
`intent_label` spellings (non-blank)	29, of which 4 cannot be mapped
Ticket questions without label	25 / 229 (10.9%): **Monday 30.0% (9/30)** vs other days 8.0% (16/199)
Message length (chars)	median 85, Q3 137.5, max 9,011 (4 pasted e-mail chains)
Personal-data patterns in raw text	e-mail 17 rows, phone 11 rows, payment card 3 rows
Non-English rows	6 (Vietnamese)
Handle time (ticket level)	222 plausible: median **13.1 min**, mean 14.17, p90 22.37; 3 zero, 2 negative, 3 > 480 min
CSAT missing	resolved 47.4%, pending 91.3%, escalated 100%
Knowledge base	14 articles: 1 empty body, 1 exact duplicate, 1 draft (not approved), 1 stale (694 days), 1 without owner

Figures: `reports/figures/chart1_label_missing_by_weekday.png`, `chart2_message_length.png`, `chart3_handle_time.png`, `chart4_tickets_per_week.png`. Each one has a written interpretation in the notebook.

4. Metadata and schema quality

- **Ticket schema.** 12 columns are present and named as documented (`docs/data_dictionary.md`). Types are loaded as text first, then validated per field: timestamps are parsed explicitly, and `handle_time_min` is converted to a number only in the outcome table. **4 schema violations:** blank `ticket_id`. **5:** unparseable timestamps. `sender_role` is valid in all 706 rows.
- **Category domains.** `channel` maps to `{email, chat, web_form}` (11 → 3, all mapped). `intent_label` maps to 8 approved intents. 4 values (`urgent`, `see notes`, `misc??`, `TBD`) remain unmapped and go to **review**.
- **Knowledge metadata.** All 7 required front-matter fields are present in 14/14 articles. One article has an empty `owner` value (KB-014). `kb_id` matches the filename in 14/14 articles.
- **Processed schema.** Defined and documented in `docs/data_dictionary.md`. Every published record carries `pipeline_version`, `source_*` lineage fields and, for messages, `split`.

5. Extraction and loading quality

- **Knowledge.** 13/14 articles extracted to non-trivial text. **KB-009 extracted to 0 characters** (extraction failure) and is quarantined, not published as an empty source. Paragraph structure is preserved (published articles have `n_paragraphs` ≥ 1).
- **Tickets.** All 706 rows were parsed by the CSV reader, including multi-line quoted text. No row was dropped by the loader: the manifest has exactly 706 message rows.
- **Fidelity check.** In the 15-record manual sample, the processed text equals the whitespace-normalised, redacted raw text in **5/5** published rows, and the UTC timestamp is correct in **5/5** (section 9).

6. Quality issues found (issue register)

#	Issue	Evidence (Observed)	Severity	Treatment	Validation after treatment
Q1	Exact duplicate rows	12 rows identical in all fields	Critical	keep first, quarantine copies (`MSG_DUPLICATE_EXACT`)	0 duplicate `message_id` in processed output
Q2	Customer re-sends	6 same-text messages in the same ticket within 30 s	Warning	review (`MSG_NEAR_DUPLICATE`)	not in processed output; listed in review queue
Q3	Blank group key	4 rows without `ticket_id`	Critical	quarantine; thread and split cannot be protected	G3: 0 tickets across splits

#	Issue	Evidence (Observed)	Severity	Treatment	Validation after treatment
Q4	Empty text	6 rows	Critical	quarantine	0 empty texts published
Q5	Mixed or invalid timestamps	96 naive, 5 invalid, 2 future	Critical (invalid/future); transform (naive)	convert to UTC (naive treated as UTC+07, *Assumed*); quarantine invalid and future	all published `created_at_utc` in ISO-Z; 5/5 manual check
Q6	Category spellings	11 channel and 29 label spellings	Transform / Warning	map via config; unmapped label → review	processed channel ∈ 3 values; intents ∈ 8 values
Q7	Missing question labels (MAR, Monday-heavy)	25 / 229 questions	Warning	review for human labelling; **no** deletion and **no** imputation	0 published questions without intent
Q8	Extreme length	4 messages > fence of 371 chars (max 9,011)	Warning	review (trim pasted chains); fence fitted on **train only**	0 published messages > 371 chars
Q9	Personal data	17 e-mail, 11 phone rows	Transform	redact to `[EMAIL]` / `[PHONE]` (18 replacements in published rows)	G2: 0 residual patterns
Q10	Payment-card numbers	3 rows	Critical	quarantine; never redact-and-keep; the text is not copied to the log	G2 = 0; quarantine log holds no text
Q11	Out-of-scope language	6 Vietnamese rows	Critical (scope)	quarantine (`MSG_OUT_OF_SCOPE_LANGUAGE`)	0 non-English published
Q12	Post-outcome fields	`resolution_status`, `csat_score`, `resolved_at`, `handle_time_min`	Leakage	removed from interactions; ticket-level table for baseline only	test: 0 such fields in interactions
Q13	Impossible or extreme handle time	3 zero, 2 negative, 3 > 480 min	Flag	`handle_time_flag`; excluded from B_o; extremes **kept and flagged**	flags 222 ok / 3 zero / 2 negative / 3 extreme
Q14	Weak labels and reference truth	wrong label on a quarantined row; generic agent reply to a specific question	Warning	agent replies not used as gold answers; label audit	audit 20/20 correct (Wilson 95% lower bound ≈ 84%)
Q15	KB: empty, duplicate, draft	KB-009, KB-011, KB-013	Critical	quarantine	9 published, all approved and unique
Q16	KB: stale or ownerless	KB-012 (694 days), KB-014 (no owner)	Warning	review; owner sign-off required	not published
Q17	CSAT missingness (MNAR/MAR)	47.4% vs 91.3% vs 100% by status	Note	not used in Week 2; any later use must report response bias	—

7. Transformations applied

All transformations are in `src/data/build_dataset.py`, with rule functions in `src/data/rules.py` and settings in `configs/pipeline_config.json`. They are conservative: every transformation is reversible through `source_row` and `source_sha256`, and none invents a value.

1. Unicode NFC normalisation and whitespace collapse of text and article paragraphs.
2. Timestamp conversion to UTC (ISO-Z); naive values interpreted as UTC+07 (**Assumed**, to be confirmed by the source owner).
3. Channel and intent normalisation through explicit mapping tables. Unmapped values are **not** guessed.
4. Redaction of e-mail and phone numbers. Payment cards are never redacted-and-kept: they are quarantined.
5. Ticket-level intent propagated from the question to the rest of the thread (`ticket_intent`).
6. Post-outcome fields moved out of the interaction records.
7. Length-outlier fence = Q3 + 3·IQR, **fitted on the train split only** (371 chars) and applied to all splits.

Not applied on purpose: no label imputation, no deletion of extreme values, no translation, no chunking or embedding.

8. Published, reviewed, quarantined and excluded records

Dataset	Raw	Published	Review	Quarantine	Check
Knowledge articles	14	9	2	3	14 = 9 + 2 + 3 ✔
Ticket messages	706	547	122	37	706 = 547 + 122 + 37 ✔
Total records	**720**	**556**	**124**	**40**	**720 = 556 + 124 + 40 ✔**

- **Quarantine (37 messages):** exact duplicates 12 (one is also Vietnamese), empty text 6, Vietnamese 5, invalid timestamp 5, missing ticket_id 4, payment data 3, future timestamp 2. **Knowledge (3):** KB-009 empty, KB-011 duplicate of KB-003, KB-013 draft.
- **Review (122 messages):** label missing 25, label unmapped 4, length outlier 4, near duplicate 6 (one also thread-held), thread held because its question is in review 61, thread without a usable question 22. **Knowledge (2):** KB-012 stale, KB-014 no owner. Review rows are in `data/quarantine/review_queue.csv` with their split already assigned, so released records go to the correct split.
- **Excluded by design (fields, not records):** the four post-outcome fields are excluded from interactions. Vietnamese messages are excluded because they are out of MVP scope, not because they are bad data.
- **Published interactions:** 547 messages (276 customer, 271 agent), 188 tickets, 188 labelled questions. Splits: train 92 tickets / 263 messages, validation 49 / 147, test 47 / 137.

Readiness metrics (Observed). Message publish rate 77.48% (**target ≥ 90%, missed**). Complete-thread rate 187/230 = 81.30% (**target ≥ 85%, missed**). Knowledge publish rate 9/14 = 64.29%. The KB covers 8/8 intents. Residual personal data: 0.

9. Manual validation performed

- **Sample 1, fidelity.** Stratified random sample, `random_state=42`: **15 ticket messages**, 5 per outcome. Results: published text faithful 5/5; UTC conversion correct 5/5; channel and label mapping correct; the quarantine reason was correct for 5/5 and the review reason was correct for 5/5.
- **Sample 2, label accuracy.** `random_state=7`: **20 published ticket questions**. 20/20 labels match the question (100%; Wilson 95% lower bound ≈ 84%).
- **Findings that changed the plan:** (a) a quarantined row carries a wrong label (`Warranty` on a double-charge message), so labels are weak; (b) agent replies can be generic and not answer the specific question, so agent replies are **not** a gold reference; (c) published questions reduce to only **26 distinct wordings** after masking order numbers, so the dataset is not representative of real phrasing.

10. Leakage risks and evaluation split

- **Future unseen data** means customer questions that arrive **after** the knowledge base and any tuning examples were frozen.
- **Split:** temporal, with **ticket** as the group unit. Train is before 1 Aug 2026 (local), validation is August, and test runs from 1 Sep to the export snapshot. A ticket is assigned by its first message, so a whole thread stays in one split. G3: **0 tickets across splits**.
- **Target leakage:** `resolution_status`, `csat_score`, `resolved_at` and `handle_time_min` are only known after the outcome, so they are excluded from interactions (test-enforced). The agent reply in the same thread is the outcome of the question, so it must never be retrievable evidence; evidence comes **only** from the knowledge base.
- **Temporal leakage:** 2 future-dated messages are quarantined. Knowledge used to answer a test question must have existed at question time. This is a condition for Week 3, because `last_reviewed` is not an effective-from date.
- **Group leakage:** 0 tickets cross splits. **1 of 186 customers** appears in two splits. This is acceptable for KB-only retrieval and must be re-checked if any stage learns from customer history.
- **Preprocessing leakage:** the only learned parameter, the length fence, is fitted on train only (371 chars). The mapping tables are fixed by domain rules, not learned from data.
- **Audit question for every candidate feature or evidence source:** *Would this information genuinely exist and be accessible at the exact moment the system must make its decision or answer?*

11. Remaining risks and limitations

1. **Synthetic data.** The results prove the controls, not real-world quality. Question diversity is low (26 wordings), so retrieval scores measured on it would be optimistic. *Unknown* for real data.
2. **Targets missed.** Publish rate 77.5% (target 90%) and complete-thread rate 81.3% (target 85%). The cause is the review queue (33 questions held), not data loss.
3. **Timezone assumption.** Naive timestamps (96 rows) are assumed to be UTC+07. If they are wrong, tickets near 1 Aug and 1 Sep could change split.
4. **Label quality** is audited on 20 items only. Labels in the source can be wrong (finding 9a).
5. **Small evaluation splits.** 47–49 tickets each. Per-intent metrics are unstable (e.g. `payment_issue` has 2 questions in validation).
6. **Heuristic detectors.** The PII and language detectors are regular-expression heuristics. They pass all tests, but they could miss unusual formats such as names or addresses, which are not detected at all.
7. **Knowledge freshness** relies on self-reported `last_reviewed` dates.

12. Final readiness decision and conditions before the next stage

All critical gates pass (G1 input integrity, G2 no residual personal data, G3 split integrity, G4 accounting, G5 knowledge coverage). The published knowledge (9 approved, unique articles covering 8/8 intents) and the 547 published interactions are traceable, reproducible and safe for a Week 3 **retrieval experiment**. The data is not fully ready, because both target metrics are missed and several assumptions are still open.

Conditions before the next stage

#	Condition	Owner	Evidence of completion
C1	Label the 33 held questions (25 missing, 4 unmapped, 4 over-length) and release their threads; rerun and reach publish rate ≥ 90% and complete-thread rate ≥ 85%	Support lead + student	new `run_manifest.json` with T1 and T2 passed
C2	Owner sign-off for KB-012 (stale) and an owner for KB-014; KB-013 stays excluded until approved; re-extract KB-009	Policy Team / Logistics	KB review recorded; rerun shows them published or still excluded with reason
C3	Confirm the timezone of naive timestamps with the helpdesk owner	Source owner	config note changed from *Assumed* to *Confirmed*
C4	Build the reviewed golden question–answer set from published **validation/test** questions; do not use agent replies as gold	Student + senior agent	golden set file with reviewer IDs

#	Condition	Owner	Evidence of completion
C5	Before any claim about real performance, rerun the same pipeline on a real, de-identified export	Data owner	new inventory and run manifest

Final status: READY WITH CONDITIONS

13. Evidence paths

Evidence	Path
Raw source inventory (hashes)	`data/raw/SOURCE_INVENTORY.csv`
Raw data (synthetic, unchanged)	`data/raw/knowledge_base/`, `data/raw/tickets/tickets_export_2026-09-30.csv`
Manifest (1 row per source record)	`data/processed/manifest.csv`
Run manifest (version, config hash, counts, gates)	`data/processed/run_manifest.json`
Processed knowledge	`data/processed/knowledge/knowledge_articles.jsonl`
Processed interactions	`data/processed/interactions/train.jsonl`, `validation.jsonl`, `test.jsonl`
Ticket outcomes (baseline only)	`data/processed/interactions/ticket_outcomes.csv`
Quarantine log	`data/quarantine/quarantine_log.csv`
Review queue	`data/quarantine/review_queue.csv`
Profiling notebook + HTML export	`notebooks/02_data_profiling.ipynb`, `notebooks/02_data_profiling.html`
Figures	`reports/figures/chart1…chart6*.png`
Pipeline and rules	`src/data/build_dataset.py`, `src/data/rules.py`, `configs/pipeline_config.json`
Inventory and data generator	`src/data/make_inventory.py`, `src/data/generate_synthetic_data.py`
Tests (21)	`tests/test_rules.py`, `tests/test_pipeline.py`
Data dictionary	`docs/data_dictionary.md`

READY WITH CONDITIONS